ARCHITECTURE DECISION RECORD

Six systems, six months

A Nigerian microfinance bank

Cloud migration strategy | 7 Rs portfolio assessment

DECISION REQUESTED
Endorse ADR-0001 and ADR-0002, and ask the compliance team to decide ADR-0003 by the end of month 1.

Sakariyau Oseni · v0.1 · 15 October 2026

Public | portfolio case study

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A

Executive summary

Retain core banking, refactor mobile banking in Lagos, keep financial records on premises

6
systems, one strategy each

6
months; every step can be undone

5 of 6
old server estates switched off

3
decision records; 1 awaits compliance

SITUATION

One ageing data centre, mobile outages at salary-day peaks, and a rule that keeps financial data on premises.

RECOMMENDATION

Retain core banking, refactor mobile banking, buy a loan platform and SaaS email, retire the audit tool, move KYC documents to S3.

EXPECTED OUTCOME

No outage through the first salary cycle, automated loan decisions, and five of six old server estates switched off by month 6.

DECISION REQUESTED · Endorse ADR-0001 and ADR-0002, and ask the compliance team to decide ADR-0003 by the end of month 1.

B

Five pressures, six requirements

1 **Financial data stays on premises**

2 **Mobile outages at salary-day peaks**

3 **Costly data centre, ageing hardware**

4 **Products must launch faster**

5 **Internal systems costly to maintain**

ID	CATEGORY	MEASURABLE STATEMENT
REQ-01	**Residency**	Accounts, ledgers, transactions and loan records stay at rest on premises
REQ-02	**Peaks**	No mobile outage through the first salary cycle after cutover
REQ-03	**Lending**	Automated decisions on complete loan applications, within a time the business sets
REQ-04	**Estate**	Old servers of five of the six systems switched off by month 6
REQ-05	**Upkeep**	No self-run email servers; a vendor-supported loan platform
REQ-06	**Delivery**	Every wave lands within six months; every step can be undone

B

Scope, assumptions and constraints

IN SCOPE

- Six systems, one strategy each
- Landing zone and Direct Connect
- A six-month plan and its risks

OUT OF SCOPE

- Replacing core banking
- Cost estimates

ASSUMPTIONS

- The on-premises rule covers data at rest (compliance to confirm)
- Targets the brief leaves open are set by the bank
- Vendors can deploy in the bank's data centre

CONSTRAINTS

- No AWS Region in Nigeria; Cape Town is 48 ms or more away and back
- Lagos Local Zone: no S3, RDS, ElastiCache, Fargate or NAT gateway
- Six months, one small IT team

Current state: everything in one ageing data centre

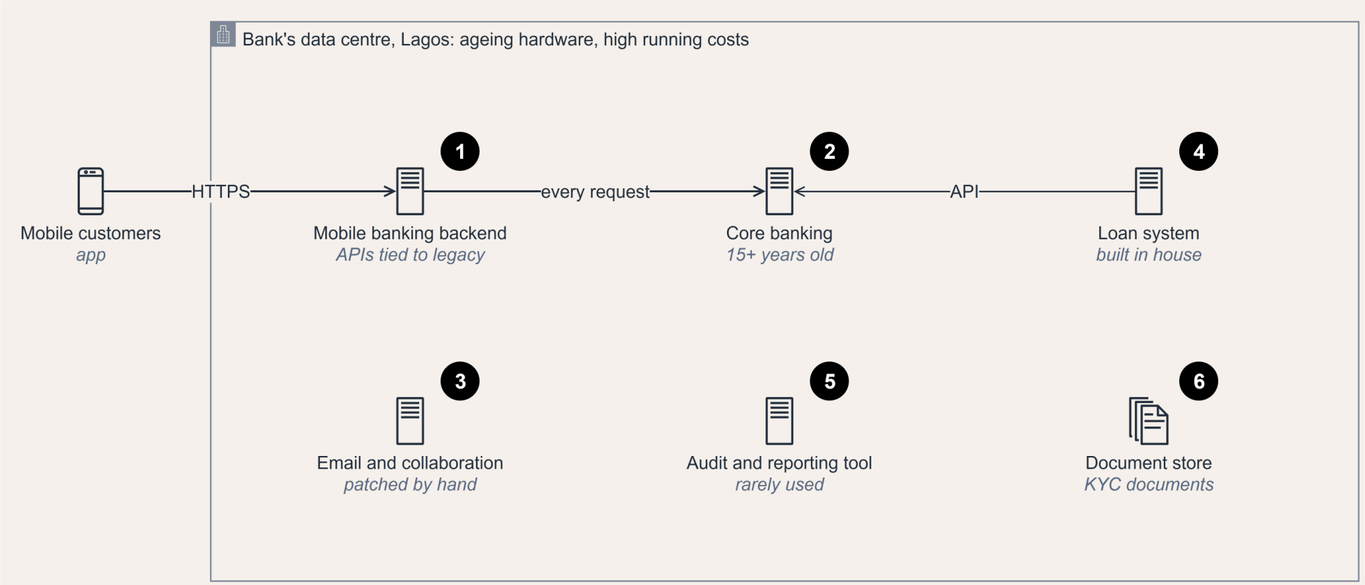


Figure 1. Today: six systems in one ageing data centre, pain points numbered

PAIN POINTS

1 Mobile backend fails at salary-day peaks; tied to legacy systems

2 Core banking: 15+ years old, wired into many systems

3 Email and collaboration patched by hand, still goes down

4 In-house loan system is slow; fintechs decide in minutes

5 Audit tool rarely used, yet still takes servers

6 KYC documents fill large, ageing storage

D

One strategy per system

SYSTEM	STRATEGY	WHY	WHERE IT LANDS
Core banking	**Retain**	The rule keeps its data on premises; too risky to move in six months	Data centre, with an API in front
Mobile banking backend	**Refactor**	Outages come from its design; stateless services scale on schedule and load	AWS Lagos Local Zone (ADR-0001)
Email and collaboration	**Repurchase**	Commodity service that eats the team's time in patching	SaaS provider
Loan processing	**Repurchase**	A proven platform gives automated decisions in months	Data centre (ADR-0002)
Audit and reporting tool	**Retire**	Rarely used; its reports already come from the BI tool	Switched off, data archived
KYC documents	**Replatform**	Storage is the problem; S3 grows without hardware	Amazon S3, Cape Town (ADR-0003)

Target state after six months

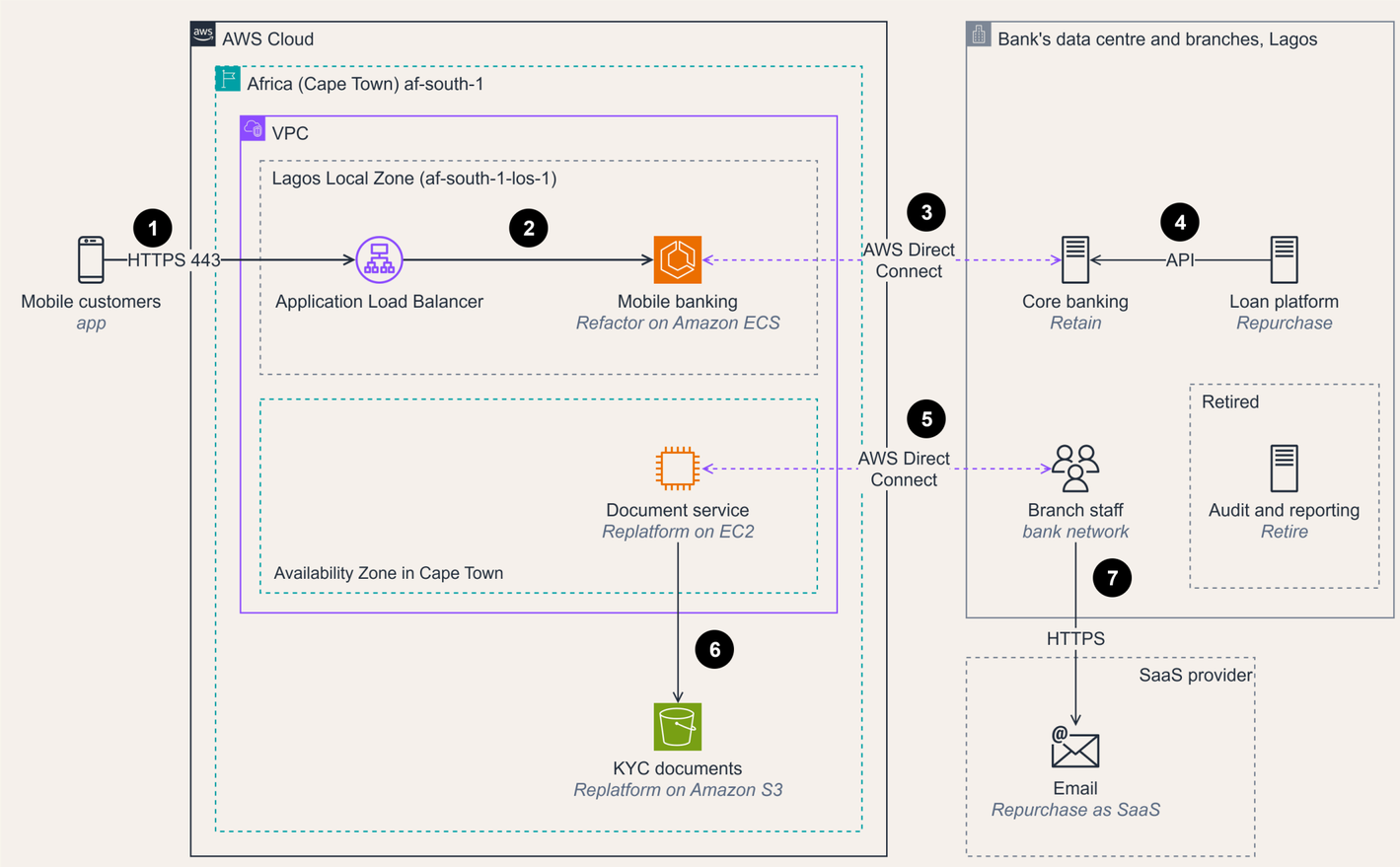


Figure 2. After six months: where each system runs, with its strategy

NOTES

1 App reaches an Application Load Balancer in the Lagos Local Zone over HTTPS

2 Refactored services on Amazon ECS scale before salary days; no records stored

3 Balances and transfers go to retained core banking over Direct Connect

4 The bought loan platform runs on premises and calls core banking's API

5 Branch staff reach the document service in Cape Town over Direct Connect

6 The replatformed document service keeps KYC documents in Amazon S3

7 Staff email and collaboration move to a SaaS provider

Salary-day payment flow

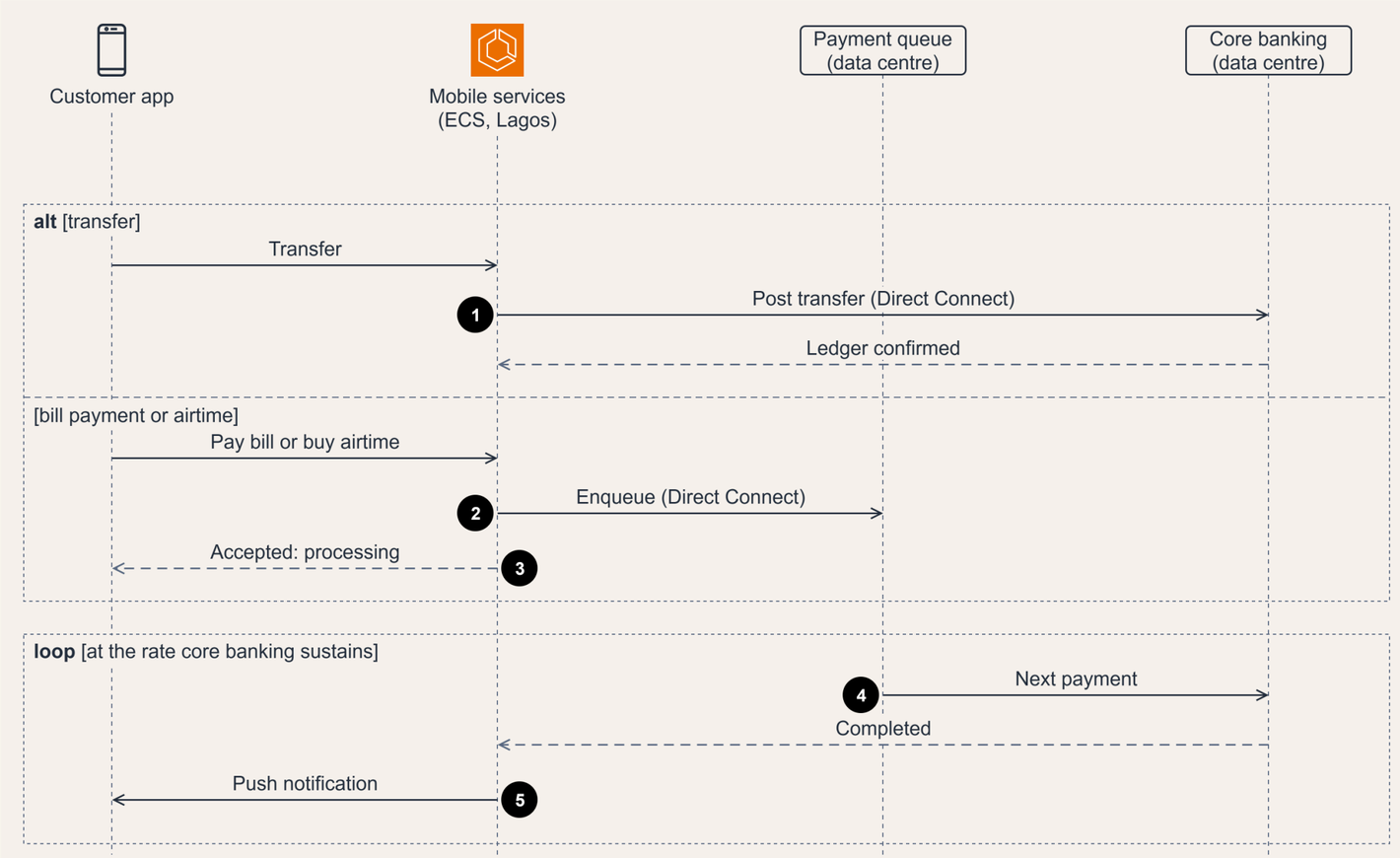


Figure 3. ADR-0001: transfers stay synchronous; payments that can wait are queued

NOTES

1 A transfer is synchronous: the service answers when the ledger confirms

2 A bill payment or airtime top-up goes into a queue in the data centre

3 The app shows it as processing at once

4 Core banking takes work from the queue at the rate it can sustain

5 The customer is notified when the payment completes

Decision log

ID	DECISION	CATEGORY	REQUIREMENTS	STATUS
ADR-0001	Refactor the mobile banking backend into the Lagos Local Zone	Compute	REQ-01, REQ-02, REQ-06	Accepted
ADR-0002	Replace the loan system with a commercial lending platform	Application	REQ-01, REQ-03, REQ-05	Accepted
ADR-0003	Move KYC documents to Amazon S3 in Cape Town	Storage	REQ-01, REQ-04	Proposed

E

ADR-0001 Refactor the mobile banking backend into the Lagos Local Zone

Accepted Confidence: Medium | Requirements: REQ-01, REQ-02, REQ-06 | Review: if AWS opens a Region in Nigeria

CONTEXT & DECISION

The mobile backend fails at peaks because it is tied to the legacy systems behind it. Core banking stays in Lagos, and Cape Town is at least 48 ms away and back.

DECISION
We will rebuild it as stateless services on Amazon ECS behind an ALB in the Lagos Local Zone, moving one route at a time (strangler fig), with payments that can wait queued in the data centre.

OPTIONS CONSIDERED

✓ Refactor in the Lagos Local Zone
Chosen: fixes the coupling, keeps calls in Lagos

Rehost on EC2 in Cape Town
Same design, 48 ms or more per call to core banking

Replatform unchanged
API tier scales, core banking stays the bottleneck

Buy servers for the peak
Idle most of the month; grows the data centre

TRADE-OFF, CONSEQUENCES & CONFIRMATION

TRADE-OFF
The most complex strategy, in a single-zone location with fewer managed services

CONSEQUENCES
+ Capacity follows demand, paid by the hour
+ Every route can move back
– Needs container and CI/CD skills
– No Fargate, ElastiCache or NAT gateway in the Local Zone

CONFIRMATION
Load test at the last salary-day peak plus a margin; no outage in the first salary cycle

E

ADR-0002 Replace the loan system with a commercial lending platform

Accepted Confidence: Medium | Requirements: REQ-01, REQ-03, REQ-05 | Review: when the regulator allows financial data in the cloud

CONTEXT & DECISION

The in-house loan system is slow and cannot scale while fintechs decide in minutes. Loan records must stay on premises, and the team is busy with the mobile refactor.

DECISION
We will buy a vendor-supported loan origination and decisioning platform, run it in the bank's data centre, and connect it to core banking and the credit bureaus.

OPTIONS CONSIDERED

✓ Commercial platform on premises
Chosen: fastest route to automated decisions

Rebuild in house, cloud-native
Slowest; competes with the mobile work

Replatform the current system
Moving it does not make decisions faster

Hosted lending service abroad
Loan records leave the data centre

TRADE-OFF, CONSEQUENCES & CONFIRMATION

TRADE-OFF
Licence fees and vendor lock-in, and less room to differentiate

CONSEQUENCES
+ Automated decisions in months
+ Upgrades and fixes move to the vendor
– Still on premises, so the data centre does not shrink
– Open loans need careful reconciliation

CONFIRMATION
Pilot branch: decision time before and after; every migrated loan matches core banking

ADR-0003 Move KYC documents to Amazon S3 in Cape Town

Proposed Confidence: Medium | Requirements: REQ-01, REQ-04 | Review: after the compliance decision

CONTEXT & DECISION

KYC documents need large, secure storage that every branch reads. Whether the on-premises rule covers them is the compliance team's call.

DECISION
If compliance approves, we will move them to an encrypted, private S3 bucket in af-south-1, copied and verified by AWS DataSync over Direct Connect, branch by branch.

OPTIONS CONSIDERED

✓ Amazon S3 in Cape Town
Proposed: durable, grows without hardware

S3 with keys in the bank's HSM
Needs 35 ms or less to the keys; Lagos is 48 ms or more

S3 on Outposts on premises
Smallest 1-year rack: 960 vCPUs, 20 TB

Retain on new hardware
Keeps the ageing-hardware problem

TRADE-OFF, CONSEQUENCES & CONFIRMATION

TRADE-OFF
The documents are stored outside Nigeria, so compliance must approve first

CONSEQUENCES
+ Designed for 99.999999999% durability
+ Private access only, over Direct Connect
− 48 ms or more per request from a branch
− Branch access depends on Direct Connect

CONFIRMATION
Compliance approval on record; DataSync reports no mismatched files

F

Risks

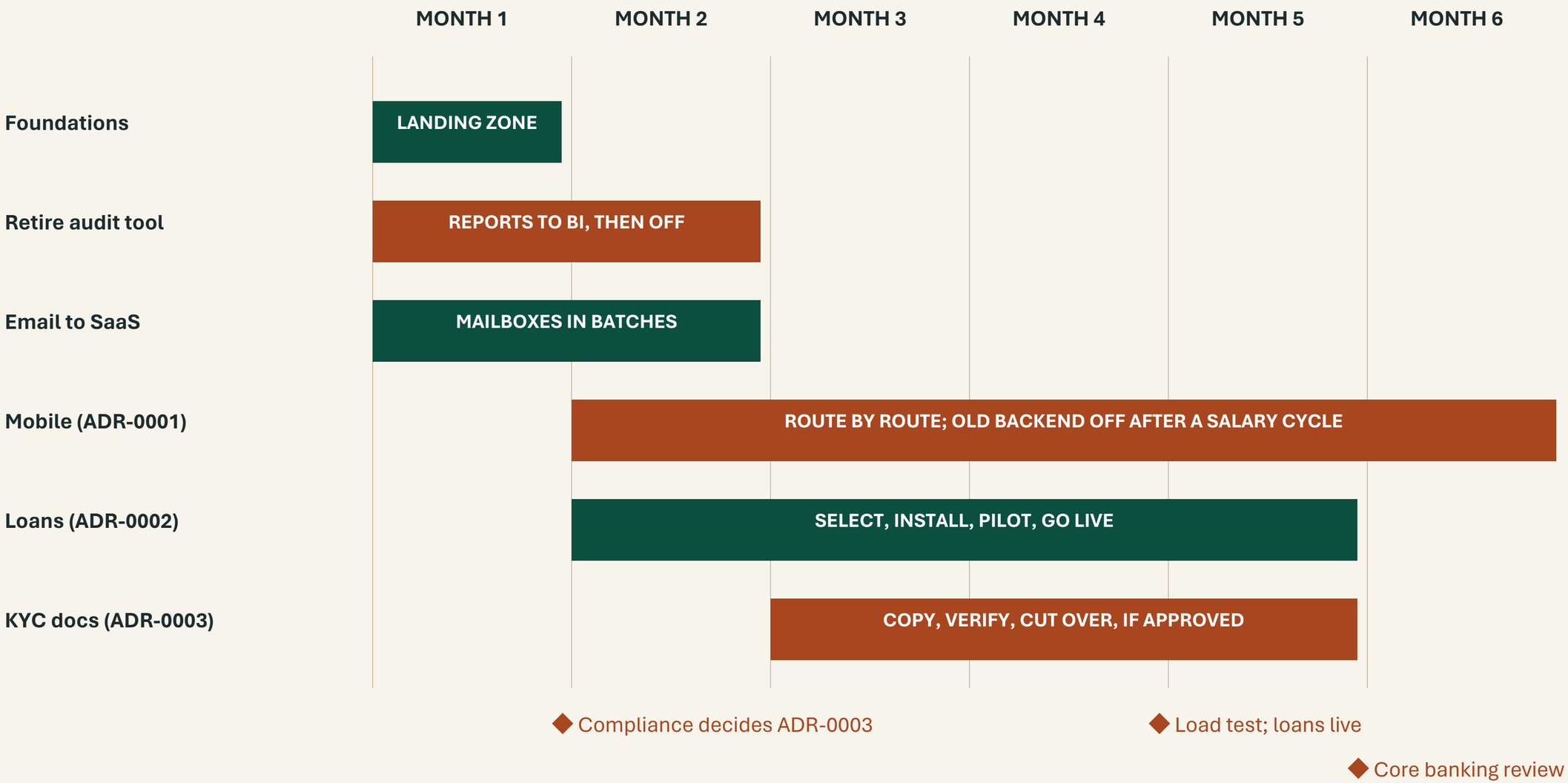
LIKELIHOOD ↑ (1 to 5)



IMPACT → (1 to 5)

ID	RISK	BAND	MITIGATION	OWNER
R1	Direct Connect fails; mobile loses core banking	**High 10**	Two links at two locations, plus VPN backup	Network lead
R2	Lagos Local Zone unavailable	**High 8**	Old backend as fallback; test before retiring	Platform lead
R3	Team lacks container and CI/CD skills	**High 12**	Training before month 2; partner on first routes	Head of IT
R4	Lock-in to the lending platform vendor	**High 9**	API-only integration; tested data-export clause	Procurement
R5	Migrated loans do not reconcile	**High 8**	Reconcile every open loan before switch-off	Operations lead
R6	Compliance keeps KYC documents on premises	**Moderate 6**	Retain on premises; supersede ADR-0003	Compliance officer

Six-month roadmap



Next steps

DECISIONS REQUESTED TODAY

Endorse ADR-0001 and ADR-0002, and ask the compliance team to decide ADR-0003 by the end of month 1.

THE NEXT 30 DAYS

1 Enable af-south-1 and build the landing zone

2 Order Direct Connect and set up the VPN backup

3 Compliance confirms the data classification and decides ADR-0003

4 List the audit reports still in use

5 Start container and CI/CD training for the team

Thank you

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github.com/ooseni/TESA-Cloud-Engineering